smalllabs

Fullstack Web Development

Session 3 | Week 3/Day 1 | 120 min

Tailwind CSS

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tailwindcss

What Is Tailwind CSS ?

What is Tailwind CSS?

Tailwind CSS is a **utility-first CSS framework** for building modern user interfaces quickly.

Instead of writing custom CSS, developers use **small utility classes directly in HTML**.

Example: Traditional CSS

```css
.button {
    background-color: blue;
    color: white;
    padding: 10px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
}
```

```html
<button class="button">Click</button>
```

Example: Tailwind CSS

```html
<button class="bg-blue-500 text-white px-4 py-2 rounded">
    Click
</button>
```

Design directly in HTML using utility classes

Utility-First Concept

What is Utility-First CSS?

A utility-first framework provides **small, single-purpose classes** that control one specific style.

Instead of writing custom CSS classes, you combine **many utility classes directly in HTML.**

Each class performs **one styling job.**

Example utilities:

- text-center
- bg-blue-500
- p-4
- rounded

These small classes combine together to create a complete design.

Traditional CSS Approach

In traditional CSS development, the styling process usually follows this workflow:

HTML → class name → CSS file → styling

Example:

```css
.button {
    background-color: blue;
    color: white;
    padding: 10px;
    border: none;
    border-radius: 6px;
    cursor: pointer;
}
```

```html
<button class="button">Click</button>
```

→

1. *Create a class name*
2. *Write custom CSS rules*
3. *The HTML references the CSS class*
4. *The CSS file controls the styling*

Problems With This Approach

- *Too many CSS files*
- *Hard to maintain large projects*
- *Class naming becomes difficult*
- *Unused CSS increases file size*
- *Switching between HTML and CSS constantly*

Tailwind CSS Approach

Tailwind changes the workflow.

Instead of writing CSS, you use **pre-built utility classes** directly in HTML.

HTML → Utility Classes → Styling

Example:

```
<button class="bg-blue-500 text-white px-4 py-2 rounded">
    Click
</button>
```

No CSS file needed.
*All styles come from **Tailwind utility classes.***

Tailwind CSS Approach

How Utility Classes Work

Each Tailwind class does **only one styling task.**

Class	Purpose	CSS Equivalent
bg-blue-500	Sets background color	background-color
text-white	Sets text color	color
px-4	Horizontal padding	padding-left & padding-right
py-2	Vertical padding	padding-top & padding-bottom
rounded	Adds rounded corners	border-radius

Installing Tailwind CSS v4

Installation Options

- CDN
- CLI
- Framework integration
  - React
  - Vue
  - Next.js
  - Laravel

Installing Tailwind CSS v4

Installing Tailwind CSS (CDN Method)

What is CDN Installation?

CDN (Content Delivery Network) allows us to **use Tailwind CSS instantly without installing anything.**

We simply add a **script link inside the HTML file.**

Installing Tailwind CSS v4

Installing Tailwind CSS (CDN Method)

Add Tailwind CDN Script

Add this line inside the <head> section of your HTML file.

```
<script src="https://cdn.tailwindcss.com"></script>
```

This method is **great for learning and quick prototypes.**

***Note:** CDN is **not recommended for production projects.**

Installing Tailwind CSS v4

Installing Tailwind CSS (CDN Method)

Example HTML Page Using Tailwind

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Tailwind CSS Button</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>

<body class="p-6">
    <button class="bg-blue-500 text-white px-4 py-2 rounded">
        Click
    </button>
</body>

</html>
```

Result

Installing Tailwind CSS v4

Installing Tailwind CSS Using NPM (Recommended for Projects)

Requirements:

- Node.js installed

```
C:\Users\lievnova>node --version
v22.19.0
```

Installing Tailwind CSS v4

Installing Tailwind CSS Using NPM (Recommended for Projects)

Requirements:

- Node.js installed

```
C:\Users\lievnova>node --version
v22.19.0
```

02 **Import Tailwind in your CSS**

Add the `@import "tailwindcss";` import to your main CSS file.

src/input.css

```
@import "tailwindcss";
```

Installing Tailwind CSS v4

Installing Tailwind CSS Using NPM (Recommended for Projects)

Requirements:

- Node.js installed

```
C:\Users\lievnova>node --version
v22.19.0
```

03

Start the Tailwind CLI build process

Run the CLI tool to scan your source files for classes and build your CSS.

Terminal

```
npx @tailwindcss/cli -i ./src/input.css -o ./src/output.css --watch
```

Installing Tailwind CSS v4

Installing Tailwind CSS Using NPM (Recommended for Projects)

Requirements:

- Node.js installed

```
C:\Users\lievnova>node --version
v22.19.0
```

04 **Start using Tailwind in your HTML**

Add your compiled CSS file to the `<head>` and start using Tailwind's utility classes to style your content.

src/index.html

```
<!doctype html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
  <link href="./output.css" rel="stylesheet">
</head>
<body>
  <h1 class="text-3xl font-bold underline">
    Hello world!
  </h1>
</body>
</html>
```

Installing Tailwind CSS v4

Installing Tailwind CSS Using NPM (Recommended for Projects)

Requirements:

- Node.js installed

```
C:\Users\lievnova>node --version
v22.19.0
```

5. Setup Script for run build CSS in *package.json*

```
{
  "scripts": {
    "build:css": "npx @tailwindcss/cli -i ./src/input.css -o ./src/output.css --watch"
  },
  "dependencies": {
    "@tailwindcss/cli": "^4.2.1",
    "tailwindcss": "^4.2.1"
  }
}
```

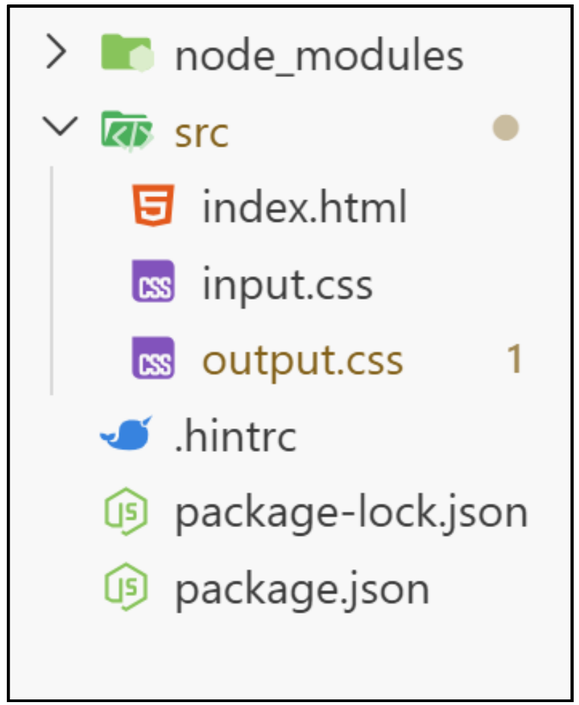
```
npm run build:css
```

6. *Run Script*

Installing Tailwind CSS v4

Tailwind Project Structure

Example:



Note: In styles.css

```
@import "tailwindcss";
```

Utility Classes

Color System

Tailwind uses color scale

Example

- blue-50
- blue-100
- blue-300
- blue-500
- blue-700
- blue-900

Example

```
<p class="text-3xl text-red-500 underline">Hello</p>
```

Utility Classes

Tailwind Spacing System

Tailwind uses a **spacing scale system.**

Spacing values are based on **rem units.**

Class	Size	CSS Value
0	0px	0
1	4px	0.25rem
2	8px	0.5rem
3	12px	0.75rem
4	16px	1rem
5	20px	1.25rem
6	24px	1.5rem
8	32px	2rem
10	40px	2.5rem
12	48px	3rem
16	64px	4rem

Utility Classes

Spacing Utilities

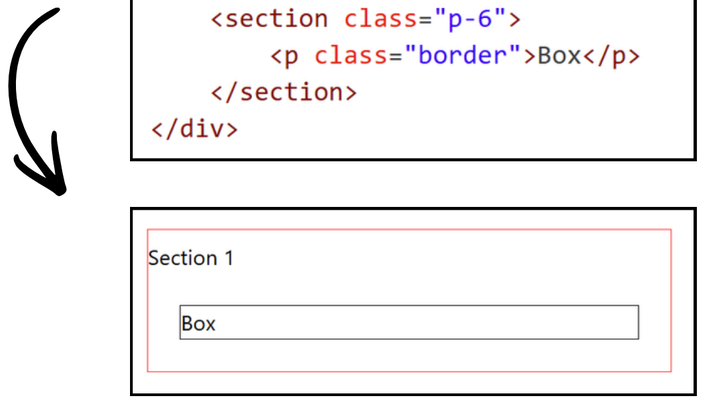
Spacing utilities control the **space inside and outside elements.**

There are **two main** types:

Type	Purpose
Padding	Space **inside** the element
Margin	Space **outside** the element

Example

```
<div class="border border-red-500">
    <section class="mt-2">
        <p>Section 1</p>
    </section>
    <section class="p-6">
        <p class="border">Box</p>
    </section>
</div>
```



Utility Classes

Padding Utilities

Padding controls **space inside the element.**

Structure: p-{size}

Padding Direction Utilities

Padding can also be applied in **specific directions.**

Class	Meaning
p-4	padding all sides
px-4	horizontal padding
py-4	vertical padding
pt-4	padding-top
pb-4	padding-bottom
pl-4	padding-left
r-4	padding-right

Utility Classes

Margin Utilities

Margin controls **space outside the element.**

Structure: m-{size}

Example

```
<div class="mt-6 mb-2 bg-red-200">
    Example Box
</div>
```

Example Box

Margin Direction Utilities

Margin can also be applied in **specific directions.**

Class	Meaning
m-4	margin all sides
mx-4	horizontal margin
my-4	vertical margin
mt-4	margin-top
mb-4	margin-bottom
ml-4	margin-left
mr-4	margin-right

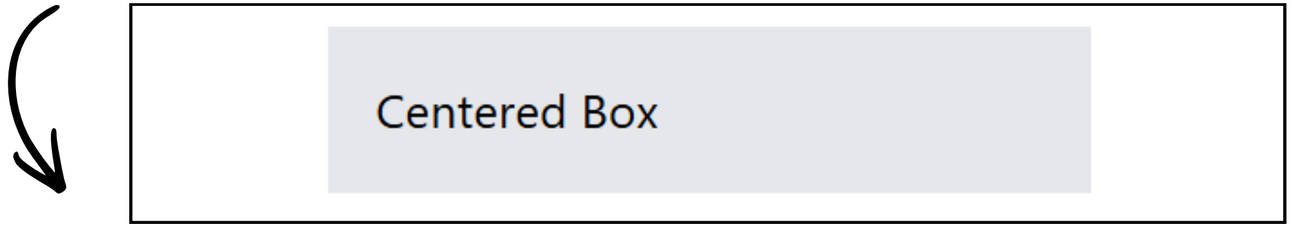
Utility Classes

Margin Utilities

- **Auto Margin :** Tailwind supports **auto margins** for layout control.

Example

```
<div class="w-64 mx-auto bg-gray-200 p-4">
    Centered Box
</div>
```



Class	Meaning
mx-auto	margin-left: auto
	margin-right: auto

This is commonly used to center elements horizontally.

Utility Classes

Space Between Elements

Tailwind provides utilities to add **space between child elements automatically.**

Structure:

- space-x-{size}
- space-y-{size}

Utility	Meaning
space-x	horizontal space between children
space-y	vertical space between children

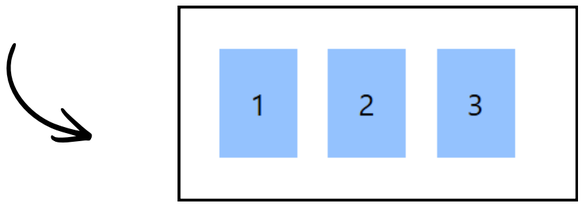
Utility Classes

Space Between Elements

Horizontal Spacing (space-x)

Example:

```
<div class="flex space-x-4">
    <div class="bg-blue-300 p-4">1</div>
    <div class="bg-blue-300 p-4">2</div>
    <div class="bg-blue-300 p-4">3</div>
</div>
```

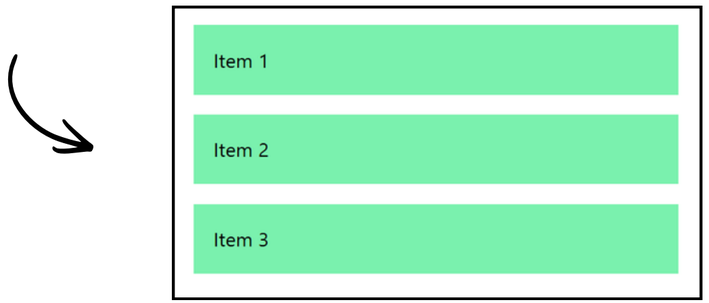


Space Between Elements

Vertical Spacing (space-y)

Example:

```
<div class="space-y-4">
    <div class="bg-green-300 p-4">Item 1</div>
    <div class="bg-green-300 p-4">Item 2</div>
    <div class="bg-green-300 p-4">Item 3</div>
</div>
```



Utility Classes

Negative Spacing

Tailwind supports negative margin values.

Example:

```
<div class="-mt-4">
    Example
</div>
```

This is useful for:

- *overlapping layouts*
- *advanced UI positioning*

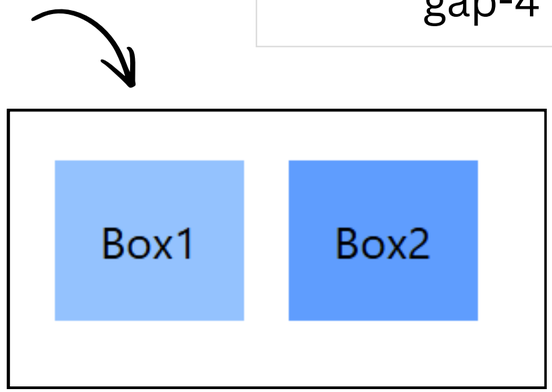
Utility Classes

Flexbox Layout

Tailwind makes layout very easy.

Example:

```
<section>
    <div class="flex gap-4">
        <div class="bg-blue-300 p-4">Box1</div>
        <div class="bg-blue-400 p-4">Box2</div>
    </div>
</section>
```



Common classes:

Class	Meaning
flex	enable flex
gap-4	space between

Utility Classes

Flexbox Layout

Flex Direction

Control direction of items:

Class	Meaning
flex-row	horizontal (default)
flex-row-reverse	reverse row
flex-col	vertical
flex-col-reverse	reverse column

Justify Content (Horizontal Alignment)

Controls alignment on main axis.

Class	Meaning
justify-start	left
justify-center	center
justify-end	right
justify-between	space between
justify-around	space around
justify-evenly	equal spacing

Utility Classes

Flexbox Layout

Align Items (Vertical Alignment)

Control direction of items:

Class	Meaning
items-start	top
items-center	center
items-end	bottom
items-stretch	stretch
items-baseline	align text

Flex Wrap

Controls whether items wrap to next line.

Class	Meaning
flex-wrap	allow wrap
flex-nowrap	no wrap
flex-wrap-reverse	reverse wrap

Utility Classes

Typography

Control text styles easily.

Example:

```
<section>
    <h1 class="text-4xl font-bold">
        Title
    </h1>
    <p class="text-gray-600 text-lg">
        Paragraph
    </p>
</section>
```

Title

Paragraph

Common classes:

Class	Meaning
text-sm	small
text-lg	large
font-bold	bold
text-center	center

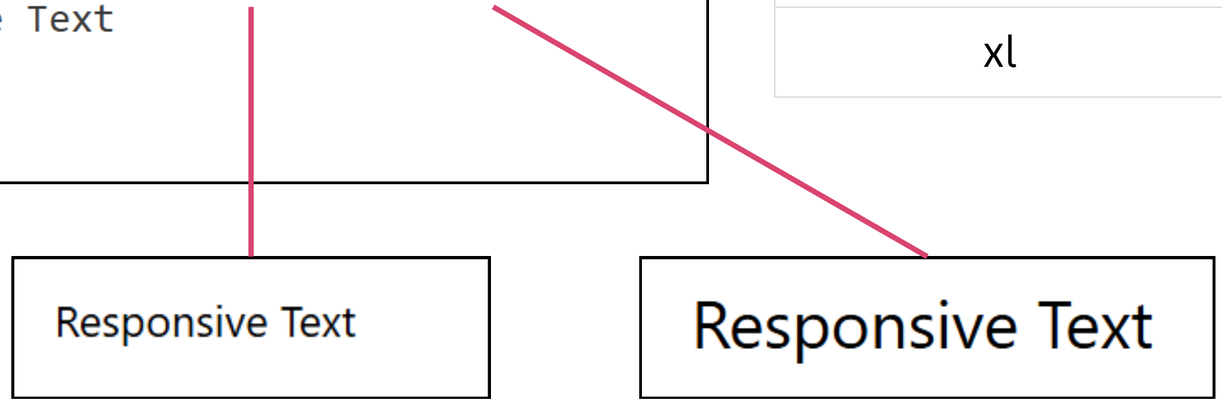
Utility Classes

Responsive Design

Tailwind uses mobile-first breakpoints.

Example:

```
<section>
    <div class="text-sm md:text-lg lg:text-2xl">
        Responsive Text
    </div>
</section>
```



Breakpoints:

Prefix	Screen
sm	≥640px
md	≥768px
lg	≥1024px
xl	≥1280px

Dark Mode

Tailwind enable dark mode easily.

Example:

```
<div class="bg-white dark:bg-gray-900 text-black dark:text-white">
    Dark Mode Example
</div>
```

Tailwind v4 Customization (@theme)

New in Tailwind v4: CSS-first configuration

Instead of config file, we use:

```
@theme {
  --color-primary: #3b82f6;
}
```

What is @theme?
It defines design tokens like:

- colors
- spacing
- fonts

Using Custom Theme

```
<div class="bg-[--color-primary] text-white p-4">
    Primary Box
</div>
```

Tailwind Layers

What are Tailwind Layers?

Tailwind organizes styles into **three main layers:**

- **Base**
- **Components**
- **Utilities**

These layers control **how styles are applied and overridden.**

Layer Order: Base → Components → Utilities

Tailwind Layers

@layer base — Default / Global Styles

The **base layer** is used for:

- Reset styles (like normalize.css)
- Default HTML styling
- Global typography

Example:

```
@layer base {
  h1 {
    font-size: 2rem;
    font-weight: bold;
  }

  body {
    font-family: sans-serif;
  }
}
```

Tailwind Layers

@layer components — UI Components

The **components layer** is used to create:

- Buttons
- Cards
- Forms
- UI blocks

Example:

```
@layer components {
  .btn-primary {
    @apply px-4 py-2 bg-blue-500 text-white rounded;
  }
}
```

```
<button class="btn-primary">
    Save
</button>
```

Tailwind Layers

@layer utilities — Custom Utility Classes

The **utilities layer** is for:

- Small helper classes
- One-purpose styles

Example:

```
@layer utilities {
  .text-shadow {
    text-shadow: 2px 2px 5px gray;
  }
}
```

THANK YOU

For your attention !

tailwindcss